

Yash Patel

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Mechatronics Engineering

Sep. 2022 – May 2027

Relevant Courses: Mechanical Engineering Design, Mechanics of Deformable Solids, Microprocessors and Digital Logic, Sensors and Instrumentation, Circuits, Structures and Properties of Materials

EXPERIENCE

Autonomous Robotics SWE Intern

Sep. 2025 – Dec. 2025

Daylun Housing

Toronto, ON

- Developed an **autonomous pick-and-place system** using **ROS2, MoveIt, and Gazebo**, writing custom joint-space controllers for stud placement onto panel frames, achieving **95%** placement accuracy in simulation.
- Engineered **closed-loop nailing, spray-foam dispensing, and steel-strapping subsystems**, integrating actuator control and sensor feedback to automate sequential panel fabrication steps.
- Simulated the **full AMR pipeline in NVIDIA IsaacSim** across **6** stations, implementing per-station **state machines** to validate end-to-end timing and correctness, reducing debugging cycles by **20%**.

Co-Founder & Erosion/Software Team Lead

Nov. 2023 – May 2025

WatDig Design Team

Austin, TX

- Designed the **motor, gearbox, and couplings** in **OnShape** to withstand the **2200 Nm** torque requirement.
- Utilized **rapid prototyping** to create enclosures and mounts for above and below ground PCB control stations.
- Developed an advanced **muck removal system** incorporating a **venturi pump** to effectively dampen and discharge soil during excavation, improving debris clearance efficiency by **35%** and reducing system downtime.
- Designed and fabricated a full **vehicle chassis** using **SolidWorks**, machined to tolerance using a **milling machine** and **lathe**, ensuring structural integrity and precise component fitment across **10+** interfacing parts.
- Engineered **mounts** and **housings** for **Ultra Wideband** sensors, batteries, and motor drivers using **SolidWorks**, consolidating **6+** components into a compact and serviceable layout within the vehicle frame.

CAD/CAM Programmer

Jan. 2025 – Apr. 2025

Sable Metal Manufacturing

Markham, ON

- Programmed sheet metal **bending sequences** using **INFORM II** for the **Motoman HP-50** robotic arm to automate production, resulting in a 15% increase of efficiency.
- Designed 3D models of sheet metal components for **laser cutting, bending, and welding** using **SolidWorks**.
- Engineered a novel **end-of-arm tooling system** incorporating a **pneumatic gripper** and suction cups to expand part-handling capabilities.
- Implemented a custom **mechanical limit switch** based on **parallel linkage mechanics** to enable seamless production across varying part geometries.

Vehicle Platform Division - Interfacing/Mechanical Team Member

Jan. 2024 – Nov. 2024

WATonomous Design Team

Waterloo, ON

- Designed and fabricated a **roof rack** in **SolidWorks** to structurally mount and expose **4 LiDAR sensors** and **3 cameras**, optimizing sensor placement for **360°** coverage.
- Engineered a compact **SolidWorks**-designed casing to house an **emergency stop** and rocker switch within a cup holder, ensuring accessible vehicle safety controls while maintaining a clean and space-efficient interior integration.

Mechanical/Manufacturing Engineering Intern

Sep. 2023 – Dec. 2023

Linamar Corporation

Guelph, ON

- Created engineering **drawings** and **models** of battery enclosures, subframes, sprockets, and wheel spindles using both **SolidWorks** and **AutoCAD**, contributing to structure and chassis development.
- Created **Excel** spreadsheets detailing part information, location, and functionality as well as creating **drawing sections** resulting in a **20%** more efficient and streamlined document retrieval process.
- Designed optimized **floor layouts** for new plant areas using **AutoCAD**, reducing part retrieval time by **15%**.

SKILLS

Computer Aided Design: SolidWorks, Inventor, AutoCAD, Revit, Onshape, SketchUp, Photoshop, Fusion360

Programming Languages: C++, Python, Excel VBA, MATLAB, SQL

Mechanical Concepts & Practices: FEA, DFMA, GD&T, PFMEA, 5S, Machining, Soldering, 3D Printing